

50.039 Theory and Practice of Deep Learning

W13S1 – End and Review

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Some admin stuff

Final exam

- Content will be everything from W1 to W12 included.
- Similar format as MidTerm.
- Details for exam (location, assignments, etc.) to be sent via email.

Regarding your project

- No extension will be given (have to give grades to OSA!)
- And want to be able to see your projects before presentations and discussions on W13.
- Check schedule for presentation timeslots.

So this is the end

What's next?

Develop a deeper understanding of...

Improving training procedures (W2++)

At the moment, mostly using gradient descent algorithms to train our models...

Many different directions have been considered, for instance:

- Using Forward-Forward (2022 proposal from Hinton, to replace our conventional backprop?): <https://arxiv.org/abs/2212.13345>
- Training AIs to train other AIs?: “learning to learn” or “meta-learning”
Curious?: <https://machinelearningmastery.com/meta-learning-in-machine-learning/>

On top of everything we have seen...

Advanced Computer Vision (W4++ and W10++)

Consider enrolling for the Computer Vision Term 7 course for more advanced concepts on CV, such as:

- More advanced computer vision tasks (image segmentation, scene understanding, etc.)
- Advanced architectures like Siamese networks, and loss functions like triplet loss, more on Vision Transformers, etc.
- More on image generation with stable diffusion, etc.
- Video data models, etc.

<https://istd.sutd.edu.sg/undergraduate/courses/50035-computer-vision>

Develop a deeper understanding of...

Advanced word embedding and NLP problems (W5++ and W8++)

Many more mechanisms when it comes to embedding and language related problems.

- E.g. more advanced embeddings, a deep dive into transformers, etc.
- Typical tasks in NLP (chatbots, context propagation, sentiment analysis, language translation, summarization, etc.)

Go for the Term 7 NLP course!

<https://istd.sutd.edu.sg/undergraduate/courses/50040-natural-language-processing>

Develop a deeper understanding of...

Advanced word embedding and NLP problems (W5++ and W8++)

And some more courses on transformers/LLMs/GenAI.

1. https://www.linkedin.com/posts/amanc_artificialintelligence-machinelearning-activity-7106125333799571457--dH5/
2. https://www.linkedin.com/posts/hoang-van-hao_stanford-university-just-released-the-full-activity-7385821073260544000-z3MH/?utm_source=share&utm_medium=member_android&rcm=ACoAAAbntaYBN_8Rx2Ty30AiacJ27hkp-JDXY-8
3. https://www.linkedin.com/posts/stasbel_if-youre-serious-about-growing-in-aiml-activity-7371553237038063616-z48T/
4. https://www.linkedin.com/posts/eric-vyacheslav-156273169_just-came-across-the-most-comprehensive-llm-activity-7146153808933400576-laP9/

Develop a deeper understanding of...

Advanced attacks and Defense mechanisms (W6++)

Many more mechanisms when it comes to attacking and defending a Neural Network, e.g. new types of attacks, such as:

- Poisoning attacks (attempt to poison the dataset so the NN cannot re-train properly, especially if online learning is used),
- Weights changes (attempt to change a small subset of the weights of the NN to prevent it from working in certain ways),
- Etc.

<https://www.comp.nus.edu.sg/~reza/courses/cs6231/>

+ AdvML elective course in MSc DAI-E Security and AI track at SUTD!

Develop a deeper understanding of...

Advanced Graph Neural Networks (W9++)

We barely scratched the surface of Graph Theory. If you need to study a new math theory, let it be graph theory!

- Good graph theory course here:

<https://ocw.mit.edu/courses/mathematics/18-217-graph-theory-and-additive-combinatorics-fall-2019/>

- More advanced problems and concepts on Graph Neural Networks in lectures 1-9 of course here:

<https://www.cs.ox.ac.uk/teaching/courses/2020-2021/advml/>

Develop a deeper understanding of...

Advanced Graph Neural Networks (W9++)

Also, keep in mind that Neural Networks are graphs...

- So technically, we could build a Neural Network, which receives another Neural Network as its input...!
- What could be the uses for such a technique?
- Meta-learning? (i.e. training an AI to train another AI?!)

<https://machinelearningmastery.com/meta-learning-in-machine-learning/>

Develop a deeper understanding of...

Advanced Generative Models (W10++)

- Advanced GANs, operating on other types of data than just images (sound, text, etc.)
- Very good online course here:
<https://cs236g.stanford.edu/>
- Did not discuss how to train a Variational AutoEncoder due to its “random/stochastic” layers. This is worth looking into.
- Also worth looking into Boltzmann Machines, as they also used a similar procedure.
- Will likely be covered in **Term 7 Computer Vision**.

Develop a deeper understanding of...

Advanced Reinforcement Learning (W11++)

Barely scratched the surface about Reinforcement Learning.

- Currently considering to create a Reinforcement Learning elective course at SUTD for Term 8. Any thoughts?
- Otherwise, the reference course on RL is the one from David **Silver** (the man behind AlphaGo!)

<https://deepmind.com/learning-resources/-introduction-reinforcement-learning-david-silver>

And <https://www.davidsilver.uk/teaching/>

+ RL elective course in MSc DAI-E Embodied AI track at SUTD!

Develop a deeper understanding of...

Advanced Interpretability, physics-informed NNs (W12++)

- A very active field in research at the moment, as people are trying to understand the limitations of LLMs and the reasons why they tend to hallucinate or fail to follow moderation rules.
- Not that many courses out there, but worth keeping an eye out...

Develop a deeper understanding of...

Advanced Diffusion Models (W12++)

- Any good course about advanced diffusion models and advanced generative models (Dall-E and MidJourney) would also be worth considering.

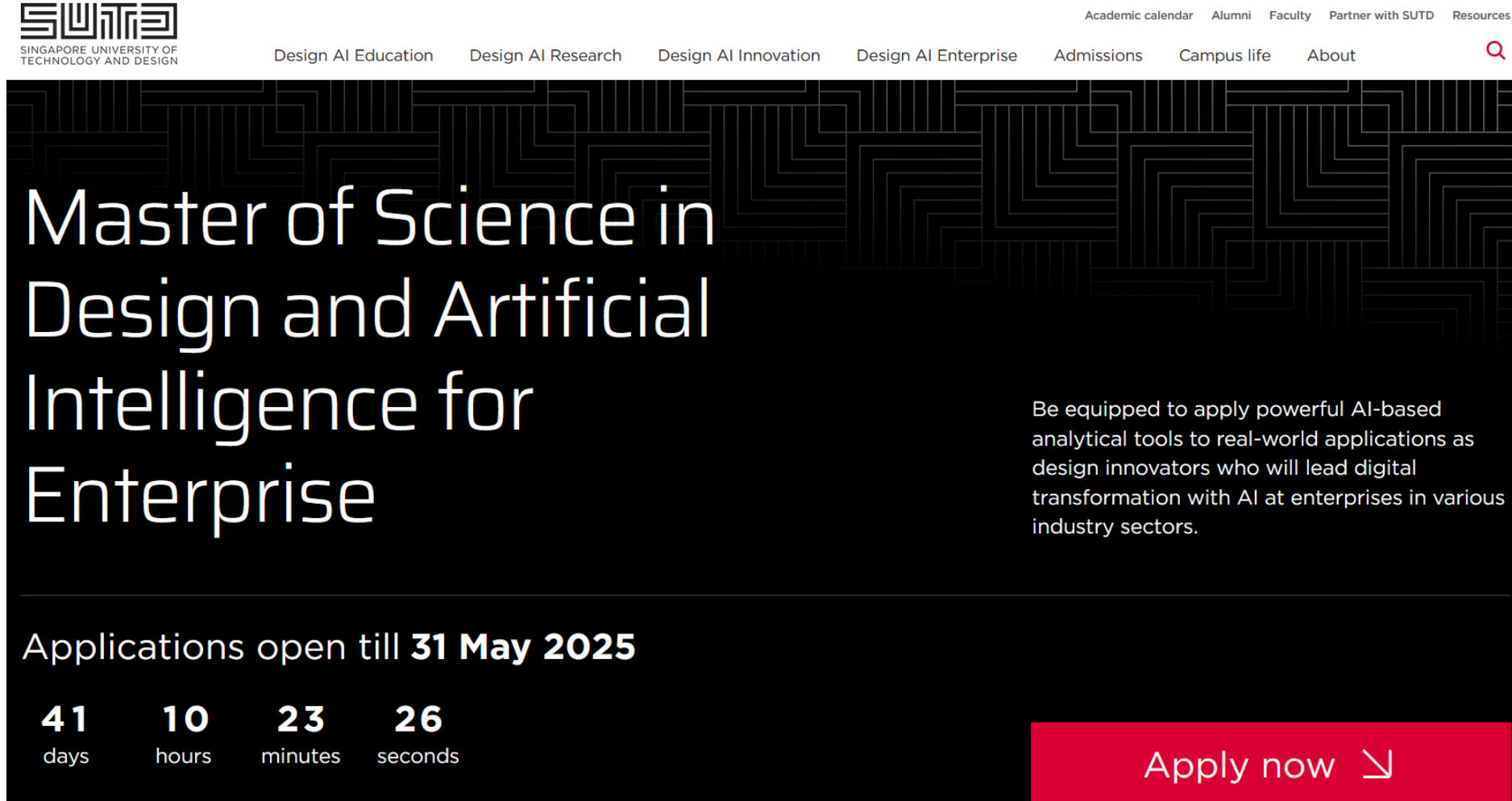
<https://www.fast.ai/posts/part2-2023.html>

- Will likely be covered in **Term 7 Computer Vision**.

Our new SUTD MSc in AI!

Open for
application:

<https://www.sutd.edu.sg/programme-listing/master-of-science-in-design-and-artificial-intelligence-for-enterprise/>



The screenshot shows the SUTD website header with the logo and navigation links: Academic calendar, Alumni, Faculty, Partner with SUTD, Resources, Design AI Education, Design AI Research, Design AI Innovation, Design AI Enterprise, Admissions, Campus life, and About. The main banner features a dark background with a white geometric pattern. The text reads: 'Master of Science in Design and Artificial Intelligence for Enterprise'. Below this, it says 'Be equipped to apply powerful AI-based analytical tools to real-world applications as design innovators who will lead digital transformation with AI at enterprises in various industry sectors.' At the bottom, it states 'Applications open till 31 May 2025' and shows a countdown timer: 41 days, 10 hours, 23 minutes, and 26 seconds. A red button with the text 'Apply now' and a right arrow is in the bottom right corner.

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Applications open till **31 May 2025**

41 days
10 hours
23 minutes
26 seconds

Apply now ↘

On top of everything we have seen...

A bit of **agentic AI** never hurts!

- A good starting point would be the 5day intensive Agentic AI course by Google: <https://www.kaggle.com/learn-guide/5-day-agents>
- DeepLearning AI/Andrew Ng courses are never a bad thing:
https://www.linkedin.com/posts/abdullah-al-noman-khu-andrew-ng-is-launching-a-new-deep-learning-activity-7387850439851094016-h6oi/?utm_source=share&utm_medium=member_android&rcm=ACoAAAbntaYBN_8Rx2Ty30AiacJ27hkp-JDXY-8
- Also, consider learning about Retrieval-Augmented Generation and **Information Retrieval** (we have a Term 7 course at SUTD, 50.045!)

On top of everything we have seen...

More concepts, problems and architectures on Computer Vision

- Also, always good to go for an image processing course to understand typical image transformation and problems out there.

https://www.coursera.org/learn/image-processing?ranMID=40328&ranEAID=*GqSdLGGurk&ranSiteID=.GqSdLGGurk-GV4LxEnPMuMd1.8y4AurRA&siteID=.GqSdLGGurk-GV4LxEnPMuMd1.8y4AurRA&utm_content=10&utm_medium=partners&utm_source=linkshare&utm_campaign=*GqSdLGGurk

On top of everything we have seen...

Bayesian and Statistical Learning (Variational AutoEncoders were 101, more on diffusion models).

- A good entry point for Bayesian Deep Learning

<https://medium.com/@ODSC/introduction-to-bayesian-deep-learning-f7568f524c90>

- Lectures 10-End

<https://www.cs.ox.ac.uk/teaching/courses/2020-2021/advml/>

On top of everything we have seen...

A bit of advanced optimization and game theory never hurts...

- Especially when trying to optimize two cooperating or competing neural networks! (GANs, actor-critic, etc.)
- Great courses here:

<https://oyc.yale.edu/economics/econ-159>

And

<https://online.stanford.edu/courses/soe-ycs0002-game-theory>

On top of everything we have seen...

CUDA masters are the king of the world these days...

- BigTech companies are looking for experts that can help with machine learning and custom GPU implementations
- The most obvious way to learn is from Nvidia courses themselves, some give certifications, but it is an investment...

<https://developer.nvidia.com/cuda-education-training>



CUDA Education & Training

Accelerate Your Applications

Learn using step-by-step instructions, video tutorials and code samples.

- Accelerated Computing with C/C++
- Accelerate Applications on GPUs with OpenACC Directives
- Accelerated Numerical Analysis Tools with GPUs
- Drop-in Acceleration on GPUs with Libraries
- GPU Accelerated Computing with Python

On top of everything we have seen...

Cloud computing is also very valuable...

- Similarly, a certification in AWS or Microsoft Azure or Google Cloud for cloud computing machine/deep learning is of high value these days!
- **In case you missed it, we have an ongoing AWS certification partnership available at SUTD!**

Details here: https://sutdapac-my.sharepoint.com/:f:/g/personal/matthieu_demari_sutd_edu_sg/lgCvSsdM8TuzQ4B1SBKLRhleAStaTQq05B32_KOp_jMSXuW?e=t9NBog



On top of everything we have seen...

Quantum is the next best thing?

Quantum computers are expected to be the next big thing in Computer Science in general.

- This will also apply to AI/ML/DL...
- This means we will get to train larger networks, faster. (This is currently a limit for many applications these days).
- Picking up on quantum computing is never a bad idea (but careful, possibly the most difficult topic out there!)

<https://towardsdatascience.com/dont-ask-what-quantum-computing-can-do-for-machine-learning-cc44feeb51e8>

<https://pennylane.ai/qml/whatisqml.html>

On top of everything we have seen...

More stuff

- Advanced Probability and Statistics (a.k.a. Statistical Learning) is always a great plus...
<https://www.statlearning.com/>
- Neuroscience should probably be part of any serious AI curriculum...
[NeuroAI] Barron et al., “What insects can tell us about the origins of consciousness”, 2015.
[Kurzweil] R. Kurzweil, “How to Create a Mind: The Secret of Human Thought Revealed”, 2012.
[Marcus] G. Marcus, “The Future of the Brain”, 2014.

Etc.

On top of everything we have seen...

More stuff

Many more courses available for free/cheap out there...

- https://www.linkedin.com/posts/endritrestelica_ai-artificialintelligence-activity-7069768595978776576-Cvsk/
- https://www.linkedin.com/posts/amanc_artificialintelligence-machinelearning-ai-activity-7052863983908753408-OPfM/

Possibly the best repo ever created on GitHub for CS stuff...

- <https://github.com/Developer-Y/cs-video-courses>

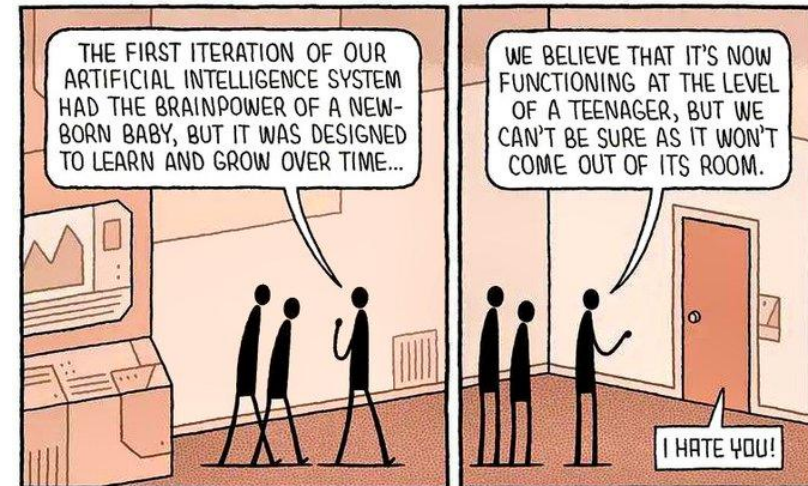
Debates about AGI

AGI: Artificial General Intelligence, naming the idea of an AI that would match all the cognitive abilities of a human.

- At the moment, definitely a dream, but the ultimate goal.
- Very active discussion topic after ChatGPT has been released.
- ChatGPT is nowhere near AGI, but is it a big step in that direction?

**I'm staying out of this debate!
(Still watching it unfold though
popcorn)**

<https://openai.com/blog/planning-for-agi-and-beyond>



TOM GAULD for NEW SCIENTIST

The important message is...

Your learning should not stop after SUTD...

Keep learning to stay up to date, this is a very fast evolving field...

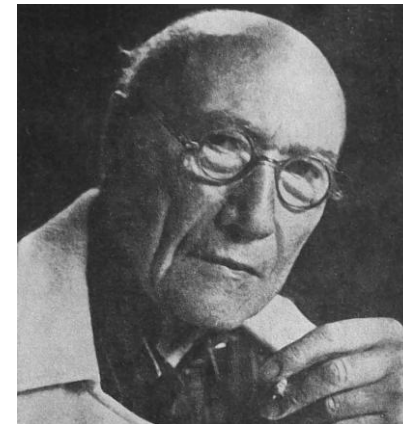
So, good luck on your continuing studies!

More importantly

DL/AI is a very active and fast-paced field.

- Keep your watchlist of papers and authors up to date.
- I have mentioned researchers, which I believe are among the most notable influencers of the Deep Learning community.
- Will be adding some more names on the next slides.

“A good professor should have this constant concern: teaching his students how to continue without him.”
– André Gide, Nobel Prize of Literature in 1947



More big names to follow

Dumping them here

Add these researchers, companies and research groups to your watchlist

- **Demis Hassabis**: Co-founder of **DeepMind**, **AlphaGo**. Several contributions in **Reinforcement Learning**.
<https://scholar.google.com/citations?hl=en&user=dYpPMQEAAAAJ>
- **Alex Graves**: Professor at **University of Toronto**. Several contributions in **Reinforcement Learning**.
<https://scholar.google.co.uk/citations?user=DaFHynwAAAAJ&hl=en>
- **Michael I. Jordan**: Professor at **UC Berkeley**, co-inventor of LDA.
<https://scholar.google.com/citations?user=yxUduqMAAAAJ&hl=fr>
- **Terrence Sejnowski**: Professor at **UC San Diego**, **Boltzmann machines**.
<https://scholar.google.ca/citations?user=m1qAiOUAAAAJ&hl=en>

Add these researchers, companies and research groups to your watchlist

- **Peter Norvig**: **Director of Research** at **Google**, co-author of the other Bible of Deep Learning
<https://scholar.google.com/citations?user=Ol0vcWgAAAAJ&hl=en>
<http://aima.cs.berkeley.edu/>
- **Stuart Russell**: **Professor** at **UC Berkely**, co-author of the other Bible of Deep Learning
<https://scholar.google.com/citations?user=2oy3OXYAAAAJ&hl=en>
- **Francois Chollet**: **Researcher** at **Google**. The man behind the **Keras** framework and **Xception**.
<https://scholar.google.com/citations?user=VfYhf2wAAAAJ&hl=en>

Add these researchers, companies and research groups to your watchlist

- **Trevor Hastie**: Professor at Stanford, co-author of the Bible of Statistical Learning.
<https://scholar.google.ca/citations?user=tQVe-fAAAAAJ&hl=en>
<https://hastie.su.domains/ElemStatLearn/download.html>
- **Robert Tibshirani**: Professor at Stanford, co-author of the Bible of Statistical Learning. Inventor of the LASSO algorithm.
https://scholar.google.ca/citations?user=ZpG_cJwAAAAAJ&hl=en
- **Vladimir Vapnik**: Retired Professor, inventor of SVMs and many other concepts. Worked with Yann LeCun at Facebook AI.
<https://scholar.google.com/citations?user=vtegaJgAAAAAJ&hl=fr>

Add these researchers, companies and research groups to your watchlist

- **Fred Cummins**: Professor at **University College Dublin**, contributions to **LSTMs** and **NLP**.
<https://scholar.google.com/citations?user=E-vg2zQAAAAJ&hl=fr>
- **Andrej Karpathy**: Former Director of AI at **Tesla**. Many contributions to Computer Vision (**Imagenet**) and NLP (**RNNs**).
(Probably better to follow him than Elon Musk.)
<https://scholar.google.com/citations?user=l8WuQJgAAAAJ&hl=fr>
- **Li Fei-Fei**: Professor at **Stanford**. Many contributions to Computer Vision (**Imagenet**).
<https://scholar.google.com/citations?user=rDfyQnIAAAAAJ&hl=fr>
- **Pieter Abbeel**: Professor at **UC Berkeley**, and a leading researcher in reinforcement learning and robotics.
<https://scholar.google.com/citations?user=vtwH6GkAAAAJ&hl=en>

Add these researchers, companies and research groups to your watchlist

- **Anil K. Jain**: Professor at **Michigan State University**. Many contributions to Computer Vision and Statistical Learning.
<https://scholar.google.com/citations?user=g-ZXGsAAAAJ&hl=fr>
- **Jitendra Malik**: Professor at **UC Berkeley**. Many contributions to Computer Vision and Statistical Learning.
<https://scholar.google.com/citations?user=oY9R5YQAAAAJ&hl=fr>
- **Sebastian Thrun**: **Stanford**, cool stuff on **robotics**.
<https://scholar.google.com/citations?user=7K34d7cAAAAJ&hl=fr>
- **Daphne Koller**: **CEO** at **InSight**, some cool **courses on Coursera**, she might be the co-founder of Coursera (?).
<https://scholar.google.com/citations?user=5lqe53IAAAAAJ&hl=en>

Add these researchers, companies and research groups to your watchlist

- **Andrew Ng: Professor at Stanford**, co-creator of **Coursera**. Has one of the best online courses on Deep Learning.
<https://scholar.google.com/citations?user=mG4imMEAAAJ&hl=en>
- **Jeremy Howard: Research Scientist at University of San Francisco**, a good scout for notable research papers on Twitter and **TED talks**.
<https://scholar.google.com/citations?user=ZWdEJ54AAAJ&hl=en>
- **Yaser S. Abu-Mostafa: Professor at CalTech**, one of the best professors for Deep Learning out there.
<https://dblp.org/pid/69/3008.html>
- **Rachel L. Thomas: University of San Francisco, FastAI**, some great TED conferences on AI and Deep Learning.
<https://scholar.google.com/citations?user=BDsAYUsAAAJ&hl=en>

Also worth subscribing to a few free newsletters about AI/DL

- The Batch newsletter by DeepLearning.ai (<https://www.deeplearning.ai/the-batch/>)
- The Algorithm by MIT Tech Review (<https://www.technologyreview.com/newsletter-preferences/>)
- The TLDR; newsletter (<https://tldr.tech/>)
- The NLP Newsletter (<https://www.ruder.io/nlp-news/>)
- The AlphaSignal newsletter (<https://alphasignal.ai/>)

Some Medium subscription never hurts (sometimes nice, easy and accessible discussions about AI).

But most seriously though, go for Twitter and follow people.

Some additional reading (from my reading list)

- <https://www.youtube.com/watch?v=-u6Ektrvk2U>
- https://www.linkedin.com/posts/amanc_artificialintelligence-machinelearning-activity-7106125333799571457--dH5
- <https://x.com/SimonPrinceAI/status/1712801009741815965?t=FDL7a74we3RzC7FTFnMXOQ&s=19>
- https://www.linkedin.com/posts/mehdi-zareb_a-visual-into-to-deep-learning-activity-7139288250514317312-mpGG/
- https://www.linkedin.com/posts/eric-vyacheslav-156273169_just-came-across-the-most-comprehensive-llm-activity-7146153808933400576-laP9/

Some additional reading (from my reading list)

- https://x.com/matdmiller/status/1743856339493757262?t=5JF_O0GEkDyAZNH4XGNOAw&s=19
- https://www.linkedin.com/posts/imarpit_ai-generativeveai-programming-activity-7240571936735399936-Epo1/
- https://www.linkedin.com/posts/fethifilali_state-of-ai-report-ugcPost-7251272468336693249-H8FS/

Let me know if you have some reading suggestions to share!

Cracking the Deep Learning interview

Most BigTech companies out there will test you on your CS/DL skills upon applying to positions/jobs in their companies.

- Train!
- Many flashcards with typical AI/ML/DL questions and tasks given in technical interviews at BigTech companies.
- <https://www.algoexpert.io/machine-learning/product>
- <https://www.mlexpert.io/>

Cracking the Deep Learning interview

More importantly, polish your GitHub with some nice ML/AI/DL projects that you can show and discuss in interviews!

- Document your codes and notebooks over the summer break!
- Many problems out there, easily investigated.
- <https://twitter.com/0xbnomial/status/1523256409529667584?t=YWXdKryCUrrJSCKWLiT4Tg&s=03>
- https://www.linkedin.com/posts/saibysani18_5-best-github-repos-to-help-you-pass-data-activity-7337825845694103552-i_9l/?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAbntaYBN_8Rx2Ty30AiacJ27hkp-JDXY-8
- Other possible idea: Find one problem for each concept we discussed each week this term and make a repo/project about it!